Plate III

The Phases of the Moon

explained by the position of the moon in its orbit

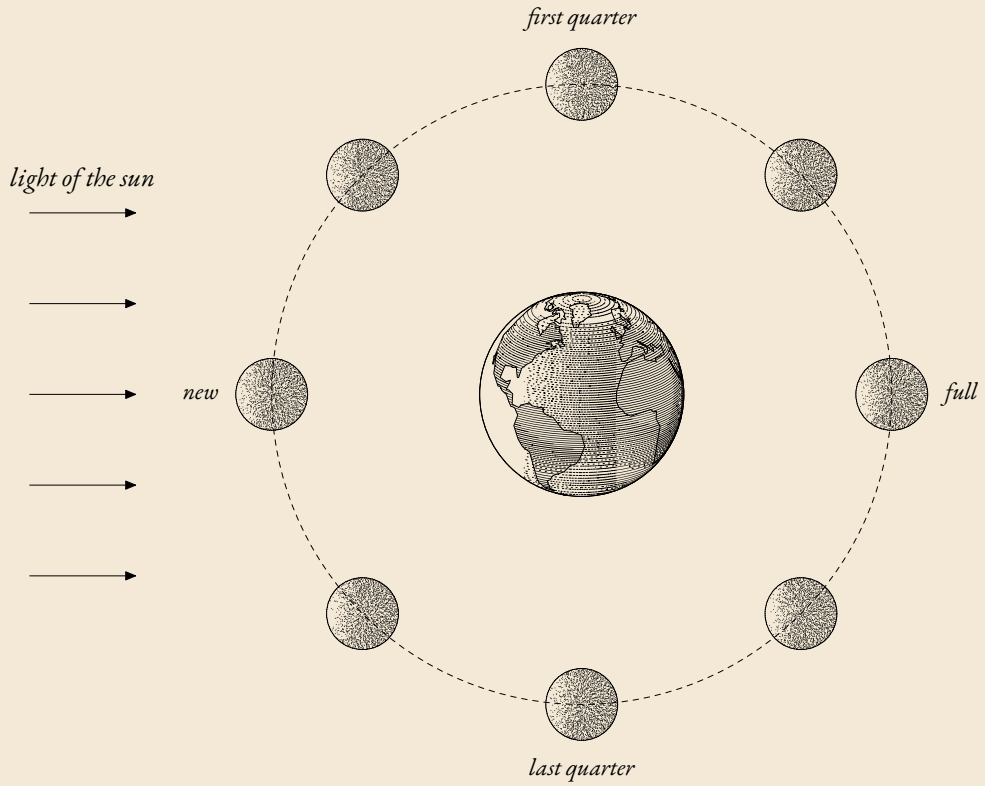


Fig. 1. The moon in eight positions of its orbit, the earth in the centre, and the sun's light coming from the left.

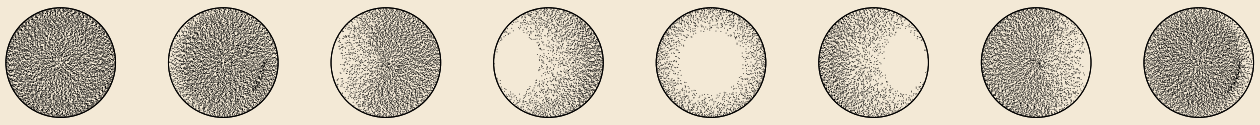


Fig. 2. The same, as seen from the earth.

The moon has no light of its own, and shows us only so much of its sunlit half as is turned toward the earth. When it stands between us and the sun that half is wholly hidden, and the moon is new; when the earth stands between, the whole of it is seen, and the moon is full. In the quarters the boundary of light and shade appears as a straight line, and in the gibbous phases it is convex, though it is at all times the edge of one and the same circle seen obliquely.

Both figures use one shading routine. In the first the light is held fixed and the moons are moved around the earth; in the second the moon is held before the observer and the light is turned about it. That the two rows agree is the whole of the explanation, and the agreement is computed rather than drawn.